

Manage storage areas

What is a storage area?

A storage area is a container where Content Platform Engine (CPE) stores content. CPE can be configured for file storage, fixed storage, database storage, or advanced storage. These storage options can be used individually or together.

When you create an object store, the wizard prompts you to specify the default content storage and the selection determines which content store serves as your default store when you add documents to the object store.

Types of storage areas

- **Database storage area**

This is the database that is used for the object store. CPE stores both the objects and the content for those objects in the same database. Database storage areas are used for a smaller number and size of documents. For larger number of documents, other storage options are preferred.

- **File storage area**

This is a storage area that contains document content in a directory tree (a hierarchy of folders) on a local or shared network drive. The disk drive can be a Windows NTFS volume, a UNIX file system, or an IBM General Parallel File System (GPFS). You cannot create a file storage area on an encrypted NTFS volume.

Content Platform Engine server must have full access to the folders and the shared network drive that is used for storage.

A many-to-many relationship exists between CPE servers and file storage areas. Many servers can manage one file storage area and a single server can manage multiple file storage areas.

File storage area contains the following directory structure:

Base directory - It is the user-named parent directory for one or more file storage areas.

Root directory - It is the user-named top-level directory for a specific file storage area. Contains a stakefile which is a system file.

Content directory tree –The directories at the lowest level of the content directory tree store the committed content element files. The Storage Area wizard creates the tree.

- **Fixed storage area**

This file storage area is an external (non-FileNet P8) fixed content system that provides more storage and data retention. It consists of a file storage staging area on the FileNet P8 system and a separate content device.

- **Advanced storage areas**

An advanced storage area supports heterogeneous storage devices. OpenStack cloud storage and file system storage, as well as IBM Cloud Object Storage (ICOS). Amazon S3, and Dell Elastic Cloud Storage can be used as advanced storage areas. Advanced storage areas provide high availability content storage and disaster recovery through the use of replication and replica repair.

If you use advanced storage areas for your object stores, you need to choose a replication model that best suits your storage requirements. Replicas are storage area devices with identical content. Advanced storage areas are designed to be flexible enough to support a wide variety of replication models. An advanced storage area replicates synchronously to a designated number of storage devices in a designated priority order. You can set up both synchronous and asynchronous replication.

Support for S3 advanced storage devices

The Content Platform Engine S3 connector provides the ability to store and retrieve documents to and from an S3-compatible object storage solution that is deployed either on premise or in a private or public cloud. The connector uses the CPE Advanced Storage Area interface that is specially designed for object and cloud connectivity.

You can use the connector with a number of devices, including:

- IBM Cloud Object Storage device
- Amazon Simple Storage Service (S3) storage device
- Dell Elastic Cloud Storage

CPE integrates with the S3 REST API and supports basic object operations such as adding, retrieving, and deleting objects in an S3 storage repository.

For more details on the supported storage solutions, review the Software Product Compatibility report for IBM FileNet Content Manager. The information is provided on the Hardware tab of the report. Reports can be generated here:

<http://www.ibm.com/software/reports/compatibility/clarity/index.html>

FileNet fix pack compatibility matrices are available here:

<http://www.ibm.com/support/docview.wss?rs=3278&uid=swg27014734>

Storage area options

Content Platform Engine(CPE) offers the following options to store the content:

- **Encryption**

CPE encrypts and decrypts content by using AES in Counter mode, a Federal Information Processing Standard (FIPS) 140-compliant algorithm.

Encryption:

- protects the confidentiality of content that you add to a storage area.
- incurs a performance cost for content upload and retrieval.

The retrieval of encrypted content relies upon information that is stored in the object store database. If that information is lost, the content cannot be retrieved. To avoid such problems, regularly back up the object store database.

- **Duplication suppression**

The suppression of duplicate content potentially reduces the storage space that is required to store content. CPE checks the existing content before adding new content to the storage area. If identical content exists, the new content is not stored separately from the existing content.

Content duplication suppression:

- incurs a performance cost for uploading content.
- does not apply to fixed content storage areas.

- **Compression**

Content that is uploaded to a storage area is compressed if content compression is enabled and if the content can be compressed below the content compression threshold.

Content compression:

- reduces storage space that is required for content storage.
- can affect overall performance.

If both compression and encryption are enabled, compression is applied before encryption. Each compressed block is encrypted independently.

- **Content Caching options**

Content caching provides faster access to content across sites by temporarily storing remote content locally.

- To reduce network traffic, content can be cached on the file system that is local to the Content Platform Engine (CPE) server.
- A content cache area is an area that contains frequently accessed document content that is duplicated from the original content in storage areas.
- The following content caching options are available:
 - Not Allowed: Storage area content caching is disabled.
 - Cross-site Only: Caching of storage area content is available only when the storage area does not belong to the same site as the server that accesses the content.
 - Allowed: Storage area content can be cached to any cache area.

Note that if encryption is enabled, the content will be encrypted in the content cache as well as in the storage area.

- **Storage level hold for Hitachi devices**

Support has been added for implementing storage level holds on content stored on Hitachi HCL storage devices.

This is an aligned mode feature:

Both the CPE software and the Hitachi storage hardware set the storage hold. CPE controls the storage hold and determines when to set and clear the storage hold on the Hitachi storage device. When a storage hold is set, content cannot be deleted until the storage hold is removed.

The storage hold capability is enabled through the Administration Console for Content Platform Engine. When enabled, all new content that is stored on Hitachi storage has a hold placed on it. When the content is to be deleted, CPE removes the hold prior to deleting the content on Hitachi storage.

Storage level hold applies to content that is ingested after the capability is enabled. You cannot apply storage holds to existing content.

Storage level holds can be applied whether the Hitachi storage device is configured in aligned or unaligned mode.

Resource statuses of File Storage Area

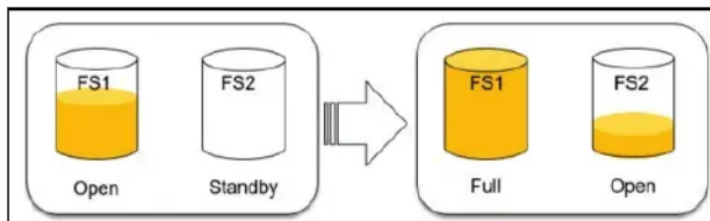
A resource status is a status associated with a storage area that determines, in combination with the storage area configuration, the permissible content operations for that storage area.

You can configure a storage area to disable the following content operations:

- Create content and Append content
- Delete content

Resource status changes occur in the following ways.

- **Automatically:** Content Platform Engine sets the resource status to Open for newly created storage areas, and also for storage areas with a status of Standby (in some circumstances). CPE changes the resource status from Open to some other status when detecting a particular storage area condition. For example, as shown in the following diagram, if FS1 reaches the maximum size and FS2 is on Standby, then FS1 switches from Open to Full, while FS2 switches from Standby to Open.



- **Indirectly:** You indirectly change the resource status when you set a storage area to be online or offline. For example, enabling the storage area to be online causes the resource status to be Open, and disabling the storage area causes the resource status to be Closed.
- **Directly:** You can directly change the resource status for a storage area.

What is storage policy?

A storage policy provides mapping to specific physical storage areas and is used to specify where content is stored for a class or object with content (for example, a document). Each storage policy can have one or more storage areas as its assigned content storage target.

Storage area farms

A storage area farm is a group of storage areas (a subset of the available storage areas) acting as a single logical target for content storage. Through this farming, Content Platform Engine provides load-balancing capabilities for content storage by transparently spreading the content elements across multiple storage areas.

The storage policy functions as both the mechanism for defining the membership of a storage area farm, and the means for assigning documents to that farm.

You can specify a single default content storage location for a document class. If you want to use storage farm capabilities, you need to use storage policies to manage the content delivery to the different storage areas.

For more information on storage management and best practices, refer to the article: <https://www.ibm.com/developerworks/data/library/techarticle/dm-1003filenetstoragemanagement/index.html>

Activity: Create a file storage area

In this activity, you create a file storage area, set it as the default storage area for the Document class, and then test it by adding a document.

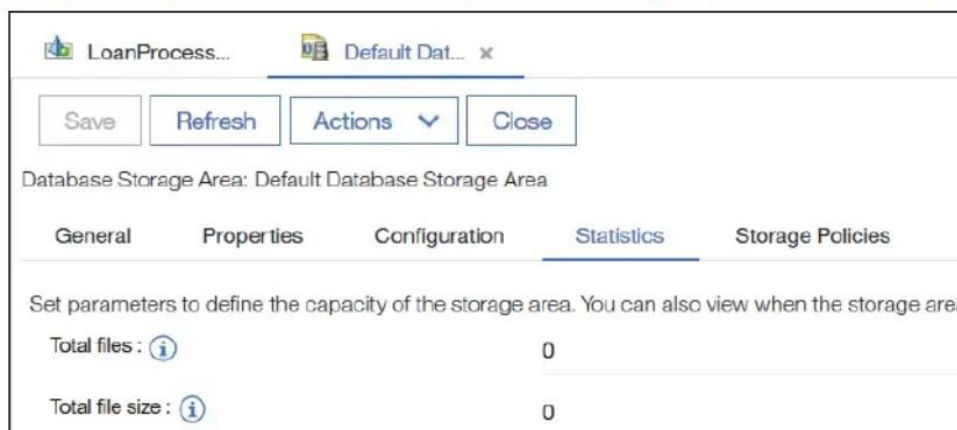
In this activity, you will accomplish the following:

- Examine the default storage area.
- Examine an existing storage directory.
- Create a subdirectory for the file storage area.
- Create a file storage area.
- Verify the storage area directory structure.
- Set default storage for the content of Document class.
- Edit your storage area.
- Add a document to verify the configuration.

Examine the default storage area.

In this task, you will add a document to an object store to check its default storage option for the Document class. You will also verify the default storage area statistics before and after adding a document.

- Ensure that the IBM FileNet P8 Platform components are started.
If you have not started them earlier, start the components by using the earlier activity: *Prepare your system - Start IBM FileNet P8 Platform*.
- In the **Mozilla Firefox** browser, click the **ACCE** bookmark or type the following URL: **http://vclassbase:9080/acce**
- Type **p8admin** for the **User name** field, **FileNet1** for the **Password** field, and then click **Log In**.
- On the left pane of the **EDU_P8** tab, expand the **Object Stores** folder and then click the **LoanProcessQA** object store.
- From the **LoanProcessQA** tab, on the left pane, expand to **LoanProcessQA > Administrative > Storage > Storage Areas** and then select **Default Database Storage Area**.
- From the **Default Database Storage Area** tab on the right pane, open the **Statistics** subtab and then click **Refresh**.
- Verify that the value for the **Total files** field is **0** (zero) and then close the tab.



- On the left pane, collapse the **Administrative** node and then expand **Browse**.
- Right-click **Root Folder** and then select **New Folder** to create a folder.
- From the **New Folder** tab on the right pane, type **Test** for the **Folder name** field and then click **Next**.
- Leave the defaults, click **Next** one more time, and then on the **Summary** page, click **Finish**.

- Click **Close** on the **Success** page and then click **Refresh** on the **LoanProcessQA** tab.
- On the left pane, expand **Browse > Root Folder**, right-click the **Test** folder, and click **New Document** to add a document.
- From the **New Document** tab on the right pane, type **TestDoc** for the **Document title** field, select the **With content** option, and then click **Next**.
- On the **Document Content Source** page, under the **Content Elements** section click **Add**.
- On the **Add Content Element** dialog box, click **Browse**.
- On the **File Upload** window, select a document (Example: **SampleDoc1.docx**) from the **C:\Training\F2810G\SampleDocs** folder and then click **Open**.
- On the **Add Content Element** dialog box, click **Add Content**.
- Click **Next** four more times (On the **Document Content Source**, **Object Properties**, **Document Content and Version**, **Specify Settings for Retaining Objects** pages).
- On the **Advanced Features** page, verify that **Default Database Storage Policy** is selected and then click **Next**.

LoanProcess... New Document... * x

< Back Next > Finish Cancel

Advanced Features

Specify additional options that you can assign to the document you are adding to the object store.

Storage area : (i)

Storage policy : (i) Default Database Storage Policy

This default policy is associated with the default storage area. After you add this document, there will be a change in the total number of files for the default storage area.

You will work with Storage policy in the following activity.

- On the **Summary** page, click **Finish** and then click **Close** on the **Success** page. View the default storage area statistics again:
- On the left pane, navigate to **Administrative > Storage > Storage Areas** and click **Default Database Storage Area**.

- From the **Default Database Storage Area** tab on the right pane, open the **Statistics** subtab and click **Refresh**.
- Verify that the **Total files** field now has a value: **1** (one).

Database Storage Area: Default Database Storage Area	
General	Properties
Configuration	Statistics
Storage Policies	
Set parameters to define the capacity of the storage area. You can also view when the storage	
Total files : 	1
Total file size : 	0

- Log out of the administration console and close the browser.

Examine an existing storage directory.

In this task, you will view the existing file storage directories.

- In **Windows Explorer**, navigate to **C:\filenet**.
The student system uses this folder as the base directory for file storage.
- Open the **filenet** folder and observe that there are several folders: **BulkMoveFS**, **file_stores**, **file_stores2**, and **PurchaseOrderFS**.

These folders are the root directories for file storage on the student system.

You can use any string value for the base and root directory names. You can select any location in your local or distributed file system for the base directory. Content Platform Engine should have full access permission to these folders.

- Open the **file_stores\content** folder and verify that there are **23** folders that are named **FN0** to **FN22**.

These directories store the committed content element files. The Storage Area wizard creates these content folders.

- Open the **FN0** (or any one of the 23 folders) folder and verify that each of them contains a set **23** folders that are named **FN0** to **FN22**.

- Open the **file_stores\inbound** folder and verify that there are several folders.

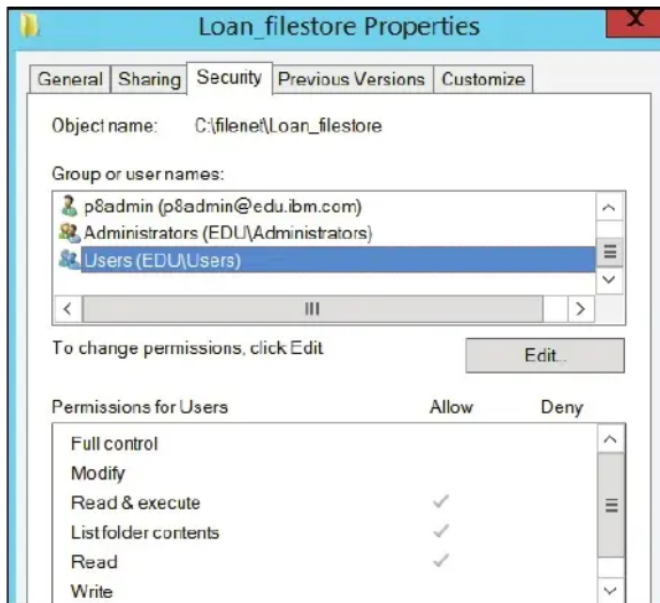
The inbound folder is the working area for uploading new content.

Create a subdirectory for the file storage area.

In this task, you will create a subfolder to use it as a root directory for the new file storage area.

- In **Windows Explorer**, navigate to the **C:\filenet** folder and then create a folder with a name: **Loan_filestore**
- Right-click the **Loan_filestore** folder and select **Properties**.

- In the **Properties** window, click the **Security** tab and then verify that the permission for the non-admin users (Example: **EDU\Users**) is **read-only** to the folder.



Only the system admin user must be able delete and write files in the file store directories.

- Click **Cancel** and then close the **Windows Explorer**.

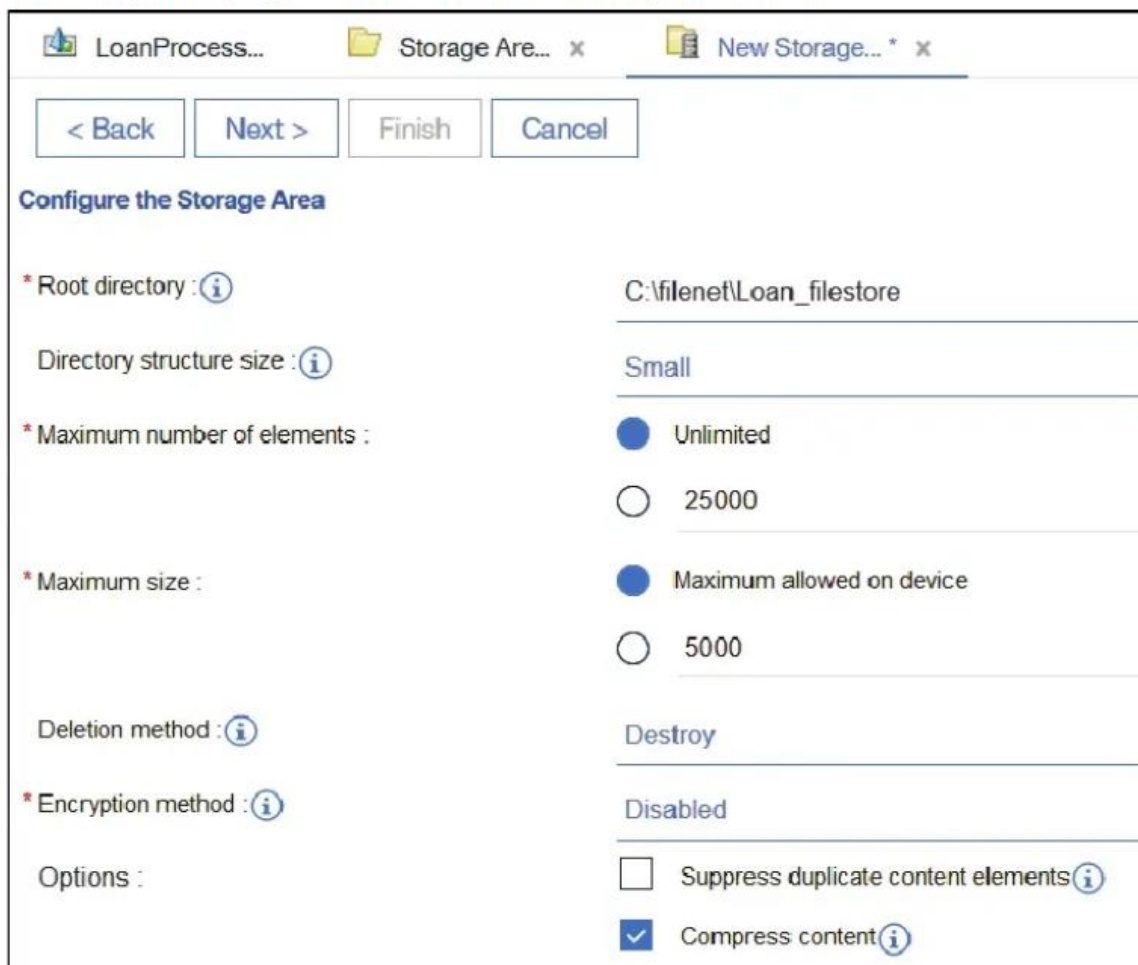
Create a file storage area.

In this task, you will create a file storage area in Administration Console for Content Platform Engine (ACCE).

- In the **Mozilla Firefox** browser, click the **ACCE** bookmark or type the following URL: **http://vclassbase:9080/acce**
- Type **p8admin** for the **User name** field, **FileNet1** for the **Password** field, and then click **Log In**.
- On the left pane, expand the **Object Stores** folder and then click the **LoanProcessQA** object store.
- From the **LoanProcessQA** tab, on the left pane, expand to **LoanProcessQA > Administrative > Storage** and click **Storage Areas**.
- From the **Storage Areas** tab on the right pane, click **New**.
- On the **New Storage Area** tab, select **File** for the **Storage area type** field and click **Next**.

- Type **Loan Storage Area** for the **Display name** field, scroll down to verify that **Initial Site** is selected for the **Site** field, and then click **Next**.
- Configure the Storage Area with the following data:
 - Root directory: **C:\filenet\Loan_filestore**
This is the directory that you created in the previous task.
 - Directory structure size: **Small**
Small structure will create two levels with a total of 529 directories. Similar to the one that you inspected in the earlier task.
 - Maximum number of elements: **Unlimited**
 - Maximum size: **Maximum allowed on device**
 - Delete method: **Destroy**
 - Encrypt content: **Disabled**
 - **Options:**
 - **Suppress duplicate content elements:** Cleared (Not selected)
 - **Compress content:** Selected

The completed page contains the values that you entered:



The screenshot shows a window titled 'New Storage...' with three tabs: 'LoanProcess...', 'Storage Are...', and 'New Storage...'. The 'New Storage...' tab is active. At the top, there are four buttons: '< Back', 'Next >', 'Finish', and 'Cancel'. Below the buttons, the title 'Configure the Storage Area' is displayed. The configuration settings are as follows:

* Root directory :	C:\filenet\Loan_filestore
Directory structure size :	Small
* Maximum number of elements :	☒ Unlimited ☐ 25000
* Maximum size :	☒ Maximum allowed on device ☐ 5000
Deletion method :	Destroy
* Encryption method :	Disabled
Options :	☐ Suppress duplicate content elements ☒ Compress content

- For all other fields, leave the defaults and click **Next**.
Click the information icon next to each field name to get more details about that field.
- For **Select a Storage Policy for this Storage Area**, leave the defaults (not selected) and click **Next**.
You will create a storage policy in the next activity.
- Click **OK** to close the message about mapping the storage area to a storage policy.
- On the **Summary** page, review the details and click **Finish**.
- On the **Success** page, click **Close** to close the tab.
- On the **Storage Areas** tab, click **Refresh** and then verify that the **Loan Storage Area** is listed.
- Notice that **Loan Storage Area** has the **Type** that you assigned (**File Storage Area**) and the **Total Files** column has zero (**0**) as the value.

	Display Name	Type	Status	Total Files	Total File Size
	Default Database Storage Area	Database Storage Area	Open	1	11.1 KB
	Loan Storage Area	File Storage Area	Open	0	0.0 KB

- Log out of the administration console and close the browser.
- Close the **Storage Areas** tab and any other tabs that are not used.

Verify the storage area directory structure.

- In **Windows Explorer**, navigate to the **C:\filenet\ Loan_filestore** folder.
- Open the **Loan_filestore** folder and observe the structure.
Verify that *content* and *inbound* directories are created. The wizard also creates a folder that is called *system* and an xml file with the name: *fn_stakefile.xml*
- Expand the **content** folder and verify that there are **23** folders that are named **FN0** to **FN22**.
- Open the **FN0** (or any one of the 23 folders) folder and verify that each of them contains a set of 23 folders that are named **FN0** to **FN22**.
- Open the **C:\filenet\ Loan_filestore\inbound** folder and verify that there are several folders.
- Close **Windows Explorer**.

Set default storage for the content of Document class.

In this task, you configure the new file storage area as the default storage container for the Document class.

- In the **Mozilla Firefox** browser, click the **ACCE** bookmark or type the following URL: **http://vclassbase:9080/acce**
- Type **p8admin** for the **User name** field, **FileNet1** for the **Password** field, and then click **Log In**.
- On the left pane of the **EDU_P8** tab, expand the **Object Stores** folder and then click the **LoanProcessQA** object store.
- From the **LoanProcessQA** object store tab, expand **Data Design > Classes** on the left pane and click **Document**.
- From the **Document** tab > **General** subtab, scroll down and then for the **Default storage area** field, select **Loan Storage Area** from the list.
- Select **<None>** from the list for the **Default storage policy** field.

Default storage area : 	Loan Storage Area
Default storage policy : 	<None>

Ensure that <None> is selected for the Default storage policy. By selecting these values, you are removing the previous choice of Default Database Storage Policy. You will now be able to test your storage area.

- Click **Save** on the toolbar and then when prompted, click **Cancel** on the **Propagate Metadata Changes** dialog box.

Depending on the configuration on this page, it affects the subclasses of the Document.

- Close the **Document** tab.
- Click **Refresh** on the **LoanProcessQA** object store tab.

Edit your storage area.

This task demonstrates how you can edit an existing storage area. You will edit the Loan Storage Area that you created earlier to modify the properties and update the Statistics tab.

- From the **LoanProcessQA** tab, on the left pane, expand **Administrative > Storage > Storage Areas** and then click **Loan Storage Area**.

- From the **Loan Storage Area** tab on the right pane, click the **Configuration** subtab and then edit the following fields.
 - Content Caching: **Not Allowed**
 - Delete method: **Purge**

* Content caching : ⓘ	Not allowed
* Deletion method : ⓘ	Purge

- From the **Loan Storage Area** tab, open the **Statistics** subtab.
- In the **Storage Area Maximums** section, change **Maximum Size** to **10 MB**.
Click the circle beside the field and change the value.

Maximum size : ⓘ	☐ Maximum allowed on device
	☒ 10

- In the **Storage Policies** subtab, observe that the Storage Area is not yet mapped to any Storage Policies.
You will create a policy in the next task.

- Leave the **Loan Storage Area** tab open for the next task.
- From the **LoanProcessQA** tab, click **Refresh**.

Add a document to verify the configuration.

In this task, you will verify that adding a document (of Document class) to the system adds the content to the new file storage area. You will also verify the default storage area statistics before and after adding a document.

The **Loan Storage Area** tab is opened from the previous task. If it is not open, open it by using the following step.

- From the **LoanProcessQA** tab, expand **Administrative > Storage > Storage Areas** and click **Loan Storage Area** tab.

- On the **Loan Storage Area** tab, click **Refresh**, open the **Statistics** subtab, and verify that there are zero files.

File Storage Area: Loan Storage Area	
General	Properties
Configuration	Statistics
Storage Policies	
Set parameters to define the capacity of the storage area. You can also view when the storage	
Total files : 	0
Total file size : 	0

- Expand **Browse > Root Folder**, right-click the **Test** folder, and then select **New Document**.
- From the **New Document** tab, type the name for the document: **Storage Area Test**
- Confirm that **Document** is selected for the **Class** field and the **With Content** option is selected.
- Click **Next** and then on the **Document Content Source** page, click **Add**.
- On the **Add Content Element** dialog box, click **Browse**.
- On the **File Upload** window, select a document (Example: **SampleTextDoc2.txt**) from the **C:\Training\F2810G\SampleDocs** folder and then click **Open**.
- On the **Add Content Element** dialog box, click **Add Content**.
- Click **Next** four more times (On the **Document Content Source**, **Object Properties**, **Document Content** and **Version**, **Specify Settings for Retaining Objects** pages).
- On the **Advanced Features** page, confirm that **Loan Storage Area** is selected and then click **Next**.
- On the **Summary** page, click **Finish** and then click **Close** on the **Success** page.
- On the left pane, click the **Test** folder under **Browse > Root Folder**.
- From the **Test** tab on the right pane, click **Refresh**, verify that your new document (**Storage Area Test**) is listed, and then click **Close**.
- On the **Loan Storage Area** tab, click **Refresh**, select the **Statistics** subtab, and then confirm that the Loan Storage area now contains one file.
The value for the Total files field shows 1 (one).
- Log out of the administration console and close the browser.

Activity: Create a storage policy

In this activity, you will create two file storage areas to represent a storage area farm. You will also create a storage policy that includes both of these storage areas and assign it to the Document class. The storage policy uses the load-balancing capabilities of the Content Platform Engine to distribute content within a storage area farm. You will then add some documents to the object store and observe the file count information in the storage areas.

In this activity, you will accomplish the following:

- Create storage area farms.
- Configure a new storage policy.
- Assign the storage policy to the Document class.
- Verify that storage area farming is working.

Create storage area farms.

In this task, you create two subdirectories for storage area farms and then create two storage areas.

- In **Windows Explorer**, navigate to **C:\filenet** folder, create two folders with the following names: **FS_Farm1** and **FS_Farm2**, and close the Windows Explorer.
- In the **Mozilla Firefox** browser, click the **ACCE** bookmark or type the following URL: **http://vclassbase:9080/acce**
- Type **p8admin** for the **User name** field, **FileNet1** for the **Password** field, and then click **Log In**.
- On the left pane, expand the **Object Stores** folder and then click the **LoanProcessQA** object store.
- From the **LoanProcessQA** tab, on the left tab, expand **Administrative > Storage**, and click the **Storage Areas** node.
- From the **Storage Areas** tab on the right pane, click **New**.
- On the **New Storage Area** tab, configure the Storage Area by using the following data:
 - Storage area type: **File**
 - Display name: **FS1**
 - Site: **Initial Site**

- Root directory: **C:\filenet\FS_Farm1**
- Directory structure size: **Small**
- Maximum number of elements: **Unlimited**
- Maximum size: **20 MB**
- Delete method: **Clear**
- Encrypt content: **Disabled**
- Suppress duplicate content elements: **Cleared** (Not selected)
- Compress content: **Selected**

For the fields that are not listed here, leave the defaults.

For more step-by-step instructions, refer to the *Create a file storage area* task in the previous activity.

- In the **Storage Areas** tab, click **Refresh** and then verify that the **FS1** Storage Area is listed and it has **0** for the **Total Files** field.
 - From the **LoanProcessQA** tab, click **Refresh**.
 - Create a second File Storage area by repeating the above steps and by using the following values:
 - Storage area type: **File**
-
- Display name: **FS2**
 - Site: **Initial Site**
 - Root directory: **C:\filenet\FS_Farm2**
 - Directory structure size: **Small**
 - Maximum number of elements: **Unlimited**
 - Maximum size: **20 MB**
 - Delete method: **Clear**
 - Encrypt content: **Disabled**
 - Suppress duplicate content elements: **Cleared** (Not selected)
 - Compress content: **Selected**

For the fields that are not listed here, leave the defaults.

- In the **Storage Areas** tab, click **Refresh** and then verify that the **FS2** Storage Area is listed and it has **0** for the **Total Files**.

	Display Name	Type	Status	Total Files
	Default Database Storage Area	Database Storage Area	Open	1
	Loan Storage Area	File Storage Area	Open	1
	FS1	File Storage Area	Open	0
	FS2	File Storage Area	Open	0

- Close the **Storage Areas** tab and then from the **LoanProcessQA** tab, click **Refresh**.

Configure a new storage policy.

In this task, you create a New Storage Policy and configure it.

- From the **LoanProcessQA** tab, expand **Administrative > Storage** and click **Storage Policies** on the left pane.
- From the **Storage Policies** tab on the right, click **New**.
- On the **New Storage Policy** tab, type **Farm Storage Policy** for the **Display name** field and then click **Next**.
- On the **Select the Content Storage Method** page, choose the **Select the storage Areas from a list** option and then click **Next**.
- For the **Storage areas** field, select **FS1** and **FS2** from the list and then click **Next**.
- On the **Summary** page, review the information, click **Finish**, and then click **Close** on the **Success** page.
- On the **Storage Policies** tab, click **Refresh** and then verify that your Storage Policy is listed.
- From the **LoanProcessQA** tab, click **Refresh**.

Assign the storage policy to the Document class.

In this task, you will remove the previously assigned storage areas and configure the storage policy for the Document class.

- From the **LoanProcessQA** object store tab, expand **Data Design > Classes** on the left pane and then click **Document** class.

- From the **Document** tab on the right pane, under the **General** subtab, scroll down to the **Default storage policy** field and then select **Farm Storage Policy** from the list.
- Select **<None>** for the **Default storage area** field.

Default storage area : 	<None>
Default storage policy : 	Farm Storage Policy

Ensure that **<None>** is selected for the Default storage area. The Farm Storage policy specifies the FS1 and FS2 storage areas to save the content for the Document class.

If both the Default storage area and the Default storage policy are set, the Default storage area setting takes precedence and the storage policy that you defined is ignored.

- In the **Document** tab, click **Save**.
- When prompted, click **Cancel** on the **Propagate Metadata Changes** dialog box.
- In the **Document** tab, click **Refresh** and then click **Close** to close the Document tab.
- Log out of the administration console and close the browser.

Verify that storage area farming is working.

To verify that the storage area farm functions, you will add some documents to the LoanProcessQA object store and then view the statistics of the FS1 and FS2 storage areas.

- In the **Mozilla Firefox** browser, click the **Sample Desktop** bookmark or enter the following URL: **http://vclassbase:9081/navigator**.
- Type **p8admin** for the **User name** field, **FileNet1** for the **Password** field, and then click **Log In**.
- From the upper right, click the down arrow next to **LoanProcess** and select **LoanProcessQA** from the list.
- On the left pane, click the **Test** folder under the **LoanProcessQA** object store.
- In **Windows Explorer**, navigate to **C:\Training\F2810G\SampleDocs**, select and drag all the files (but not the subfolders), and drop them to the IBM Content Navigator **Test** folder.

In Content Navigator desktop, the Add Documents page opens.

- In **Content Navigator** desktop, click **Add** to add all the files and wait for the upload to complete.
- Verify that the documents are listed in the **Test** folder and then log out of **Content Navigator** desktop and close the browser.
- In the **Mozilla Firefox** browser, click the **ACCE** bookmark or type the following URL: **http://vclassbase:9080/acce**
- Type **p8admin** for the **User name** field, **FileNet1** for the **Password** field, and then click **Log In**.
- On the left pane, expand the **Object Stores** folder and then click the **LoanProcessQA** object store.
- From the **LoanProcessQA** tab, on the left pane, expand **Administrative > Storage**, and then click the **Storage Areas** node.
- In the **Storage Areas** tab, confirm that **FS1** and **FS2** have some documents.

Your storage policy used the load-balancing capabilities of the Content Platform Engine to distribute content within the storage area farm. The documents were added to both the file storage areas in the storage area farm.

	Display Name	Type	Status	Total Files	Total File Size
	 Default Database Storage Area	Database Storage Area	Open	1	11.1 KB
	 Loan Storage Area	File Storage Area	Open	1	0.1 KB
	 FS1	File Storage Area	Open	7	892.8 KB
	 FS2	File Storage Area	Open	3	201.0 KB

The screen capture that is shown here is a sample. Depending on the number of documents that you added and how they were load-balanced, you might get different numbers for FS1 and FS2.

- Log out of the administration console and close the browser.

Activity: Create an advanced storage area

In this activity, you will create an advanced file storage area with two replication devices: ASFD1 and ASFD2. You want to make ASFD1 the primary synchronous device, and ASFD2 the secondary synchronous device. You will then add them to the Farm storage policy.

In this activity, you will accomplish the following:

- Create Advanced Storage Devices.
- Create an Advanced Storage Area.
- Configure the storage devices.
- Edit the storage policy.
- Test the advanced storage area.

Create Advanced Storage Devices.

You must create an advanced storage device before you can use an advanced storage area.

- In **Windows Explorer**, navigate to **C:\filenet** folder, create two folders with the following names: **ADVS1** and **ADVS2**, and close the Windows Explorer.
- In the **Mozilla Firefox** browser, click the **ACCE** bookmark or type the following URL: **http://vclassbase:9080/acce**
- Type **p8admin** for the **User name** field, **FileNet1** for the **Password** field, and then click **Log In**.
- On the left pane, expand the **Object Stores** folder and then click the **LoanProcessQA** object store.
- From the **LoanProcessQA** tab, on the left tab, expand **Administrative > Storage > Advanced Storage > Advanced Storage Devices** and click **File System Storage Devices**.
- From the **File System Storage Devices** tab on the right pane, click **New**.
- On the **New File System Storage Device** tab, type **AFSD1** for the **Display name** field and then click **Next**.
- On the **Configure the File System Device** page, type **C:\filenet\ADVS1** for the **Root directory path** field and then click **Next**.
- On the **Summary** page, verify the details, click **Finish**, and then click **Close** on the **Success** page.

- On the **File System Storage Devices** tab, click **Refresh** and then verify that **AFSD1** is listed.
- From the **LoanProcessQA** tab, click **Refresh**.
- Create another File System Storage Device by repeating the above steps and by using the following values.
 - Display name: **AFSD2**
 - Root directory path: **C:\filenet\ADVS2**
- On the **File System Storage Devices** tab, click **Refresh**, verify that **AFSD2** is listed, and then close the tab.

	Display Name	Descriptive Text	Root Directory Path
	AFSD1	AFSD1	C:\filenet\ADVS1
	AFSD2	AFSD2	C:\filenet\ADVS2

- From the **LoanProcessQA** tab, click **Refresh**.

Create an Advanced Storage Area.

In this task, you will create an advanced storage area by using the two advanced storage devices that you created.

- From the **LoanProcessQA** tab, on the left pane, expand the **Administrative > Storage > Advanced Storage node** and then click **Advanced Storage Areas**.
- From the **Advanced Storage Areas** tab, on the right pane, click **New**.
- On the **New Advanced Storage Area** tab, type **ADV_SA** for the **Display name** field, verify that Initial Site is selected for the **Site** field, and then click **Next**.
- On the **Configure the Advanced Storage Area** page, select **AES Counter Mode with 128-bit key** for the **Encryption method** field and then select the **Compress content** option.

* Encryption method : 	AES Counter Mode with 128-bit key
Options :	☐ Suppress duplicate content elements  ☒ Compress content 

- Click **Next** and then on the **Associate a Storage Device with this Advanced Storage Area** page, type **2** for the **Required synchronous devices** field.

- For the **Available storage replication devices** field, select the two devices that you created: **AFSD1**, **AFSD2** and then click **Next**.
The number of required synchronous devices must be greater than zero and equal to or less than the number of available storage replication devices.
- On the **Advanced Storage Area Parameters** page, select **Validate on creation**, verify that **Auto repair on content validation** is already selected, and then click **Next**.
- On the **Select Storage Policies** page, select **Farm Storage Policy** and then click **Next**.
- In the **Summary** page, review the details and click **Finish**.
- In the **Success** page, click **Close** to close the tab.
- In the **Advanced Storage Areas** tab, click **Refresh** and verify that the **ADV_SA** is listed.

Configure the storage devices.

In this task, you will change one of the storage devices to be the primary synchronous device. The storage device settings are configured with the defaults. You can change these settings on the Devices tab of the advanced storage area.

- In the **Advanced Storage Areas** tab, click the advanced storage area that you created (**ADV_SA**).
- In the **ADV_SA** tab, open the **Devices** subtab.
- Scroll down, for the **AFSD1** row, select **Primary synchronous** from the list, and then verify that the **AFSD2** row has the value: **Secondary synchronous**.

	Device Replica Name	Device Replica Site	Device Replica Type	Deletion Method Supported	Default Synch Type
☐	AFSD1	Initial Site	File System Storage Device	Purge	Primary synchronous ▼
☐	AFSD2	Initial Site	File System Storage Device	Purge	Secondary synchronou ▼

- Click **Save** and then click **Close**.

Edit the storage policy.

The Farm Storage Policy has three storage areas that are associated with it. You associated 2 of them when you created the storage area farms and the third one for the advanced storage area. You will remove the first two file storage areas to test the third one.

- From the **LoanProcessQA** tab, on the left pane, expand the **Administrative > Storage > Storage Policies** node and then click **Farm Storage Policy**.
- From the **Farm Storage Policy** tab on the right pane, scroll down, under the **Associated Storage Areas** section, select **FS1** and **FS2**, and then click **Remove**.
- Click **Save** to save the changes.
- Log out of the administration console and close the browser.

Test the advanced storage area.

In this task, you will verify that adding a document (of Document class) to the system adds the content to the new file storage area.

- In the **Mozilla Firefox** browser, click the **Sample Desktop** bookmark or enter the following URL: **http://vclassbase:9081/navigator**.
- Type **p8admin** for the **User name** field, **FileNet1** for the **Password** field, and then click **Log In**.
- From the upper right, click the down arrow next to **LoanProcess** and select **LoanProcessQA** from the list.
- On the left pane, click the **Test** folder under the **LoanProcessQA** object store.
- In **Windows Explorer**, navigate to **C:\Training\F2810G\SampleDocs**, select and drag all the files (not the subfolders), and drop them to the IBM Content Navigator **Test** folder.

In the Sample Desktop, the Add Documents page opens.

- In **Content Navigator** desktop, click **Add** to add all the files and wait for the upload to complete.
- Verify that the documents are listed in the **Test** folder and then log out of ICN **Sample Desktop** and close the browser.
- In the **Mozilla Firefox** browser, click the **ACCE** bookmark or type the following URL: **http://vclassbase:9080/acce**
- Type **p8admin** for the **User name** field, **FileNet1** for the **Password** field, and then click **Log In**.

- On the left pane, expand the **Object Stores** folder and then click the **LoanProcessQA** object store.
- From the **LoanProcessQA** tab, on the left pane, expand **Administrative > Storage > Advanced Storage** node, and then click **Advanced Storage Areas**.
- From the **Advanced Storage Areas** tab, on the right pane, verify that **ADV_SA** has some documents.

	Display Name 	Status	Total Files	Total File Size	Delete Method	Synchronous Replicas Required
	ADV_SA	Open	10	1.7 MB	Clear	2

The Total Files column shows the number of documents that you added. The number depends on the number of documents that you added.

- Log out of the administration console and close the browser.